

Name: Answers.
 Total Marks: 20 marks
 Total Time: 20 minutes

Show working and answers on this sheet. Show working in sufficient detail to support your answers. Incorrect answers without supporting reasoning may be allocated zero marks.

Question 1**2 marks**

Match the definition on the left-hand side with the correct answer on the right-hand side by drawing a line between each one:

Recurring Decimal	highest common factor
LCM	15.068
mixed number	a real number that can be written as a ratio of two integers or a fraction excluding zero as a denominator
Rational number	$4.72\overline{59}$
HCF	least common multiple
Terminating Decimal	number written as a whole number and a fraction

Question 2**2 marks**

- a) The number of drops of water in a rectangular water feature measuring 12 kilometres by 270 metres and 38 metres deep is 123 120 000 000 000.

Express the number of drops of water using scientific notation.

$$1.2312 \times 10^{14}$$

- b) Write 6.723×10^{-5} as a normal number.

$$0.00006723$$

Question 3**2 marks**

Write each of the following in simplest index form:

a. $(15^4)^7$

$$15^{28}$$

b. $2an \times 3n \times am \times m$

$$6a^2nm^2$$

c. $\frac{6a^2}{2ab} = \frac{3a}{b}$

d. $(7 \times 5)^5$

$$7^5 \times 5^5$$

Question 4**2 marks**

Find the simple interest paid when \$20 000 is invested at the rate of 7% p.a. for two years.

$$1400 \times 2 = 2800$$

Question 5**2 marks**

George bought 35% of 80 licorice allsorts. How many did he buy?

$$28$$

Question 6**3 marks**

Find the first quantity as a percentage of the second.

a. 7 out of 50

$$14\%$$

b. 1cm out of 2m

$$0.5\%$$

c. 90 m out of 1.5 km

$$\frac{90}{1500} = \frac{9}{150} = \frac{6}{100} = 6\%$$

Question 7**2 marks**

A pair of jeans is normally \$120. They are currently marked as having a 15% discount. How much would you now pay?

$$\$18 \quad 120 - 18 = \$102.$$

Question 8**3 marks**

Evaluate the following:

a. $10^{-3} =$
 $\frac{1}{1000}$

b. $2^3 = 8$

c. $\sqrt[3]{64} = 4$

d. $\sqrt{144} = 12$
Wahh

e. 29^0
 1

f. 47^1
 47

Question 9**2 marks**

Evaluate the following, expressing answers in scientific notation.

a. $2 \times 10^{-6} \times 7 \times 10^{15} =$

$$14 \times 10^9 \quad 1.4 \times 10^{10} \text{ ①}$$

$(\frac{1}{2})$

b. $\frac{2 \times 10^{18}}{4 \times 10^{13}} =$ 0.5×10^5

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